GlyGen: computational and informatics resources for glycoscience

Search within citing articles

Integrated multiomics reveals alterations in paucimannose and complex type N-Glycans in cardiac tissue of patients with COVID-19

SP Subramanian, M Wojtkiewicz, F Yu… - Molecular & Cellular …, 2025 - mcponline.org

Coronavirus infectious disease of 2019 (COVID-19) can lead to cardiac complications, yet the molecular mechanisms driving these effects remain unclear. Protein glycosylation is …

☆ Save Cite Cited by 5 Related articles All 8 versions

Cell-free synthetic glycobiology: designing and engineering glycomolecules outside of living cells

T Jaroentomeechai, MN Taw, M Li, A Aquino… - Frontiers in …, 2020 - frontiersin.org

Glycans and glycosylated biomolecules are directly involved in almost every biological process as well as the etiology of most major diseases. Hence, glycoscience knowledge is …

☆ Save Cite Cited by 47 Related articles All 10 versions »

Glycoproteoform profiles of individual patients' plasma alpha-1-antichymotrypsin are unique and extensively remodeled following a septic episode

T Čaval, YH Lin, M Varkila, KR Reiding… - Frontiers in …, 2021 - frontiersin.org

Sepsis and septic shock remain the leading causes of death in intensive care units (ICUs), yet the pathogenesis originating from the inflammatory response during sepsis remains …

☆ Save Cite Cited by 42 Related articles All 8 versions »

Decoding the Hexosamine Biosynthesis Pathway: Implications for Novel Therapeutic Strategies in Sarcoma

P Rahimizadeh, R Miallot, C De Bellis… - Journal of Cellular …, 2026 - Wiley Online Library

Cancer cells rewire their metabolism to sustain a high proliferation rate. Sensing external cues is essential to match the metabolic fluxes of the cells to the external stimuli. As part of …

☆ Save Cite Cited by 1 Related articles All 4 versions

GlycoEnzOnto: a GlycoEnzyme pathway and molecular function ontology

T Groth, AD Diehl, R Gunawan… - Bioinformatics, 2022 - academic.oup.com

Motivation The 'glycoEnzymes' include a set of proteins having related enzymatic, metabolic, transport, structural and cofactor functions. Currently, there is no established ontology to …

☆ Save Cite Cited by 29 Related articles All 13 versions

A Strategy to Achieve Sub-Parts-per-Million Mass Measurement Accuracy of *N*-Linked Glycans Using Infrared Matrix-Assisted Laser Desorption Electrospray …

AA Adepoju, TV Palomino, AL Sohn… - Journal of Mass …, 2025 - Wiley Online Library

Mass calibration techniques are vital in achieving high mass measurement accuracy (MMA) of large biomolecules. Variable ion populations that shift the axial frequencies due to space …

☆ Save Cite Cited by 3 Related articles All 5 versions

BiomarkerKB: An integrated knowledgebase supporting biomarker-centric exploration of biomedical data

D Masood, M Kim, J Vora, R Kahsay, P McNeely, S Kim… - Patterns, 2026 - cell.com

Biomarkers are essential tools for disease detection, risk assessment, therapeutic monitoring, and precision medicine. However, biomarker data are dispersed across …

☆ Save Cite Cited by 1 Related articles All 4 versions

Maximizing glycoproteomics results through an integrated parallel accumulation serial fragmentation workflow

M Baerenfaenger, MA Post, F Zijlstra… - Analytical …, 2024 - pmc.ncbi.nlm.nih.gov

Glycoproteins play important roles in numerous physiological processes and are often implicated in disease. Analysis of site-specific protein glycobiology through glycoproteomics …

☆ Save Cite Cited by 11 Related articles All 8 versions »

GlyCompute: towards the automated analysis of protein N-linked glycosylation kinetics via an open-source computational framework

K Flevaris, P Kotidis, C Kontoravdi - Analytical and Bioanalytical Chemistry, 2025 - Springer

Understanding the complex biosynthetic pathways of glycosylation is crucial for the expanding field of glycosciences. Computer-aided glycosylation analysis has greatly …

☆ Save Cite Cited by 9 Related articles All 10 versions

The CFDE Workbench: integrating metadata and processed data from Common Fund programs

JE Evangelista, DJB Clarke, AI Byrd… - Journal of Molecular …, 2026 - Elsevier

Abstract The NIH Common Fund Data Ecosystem (CFDE) program was established to facilitate data accessibility and interoperability across multiple Common Fund (CF) …

☆ Save Cite Cited by 6 Related articles All 5 versions